Before using this Quick Reference, read
and observe the detailed Rolls-Royce Solutions
overall documentation on BlueVision!
In particular, all safety instructions
in the overall documentation must be observed!

Third-party remote control

Connection of an external remote control (shipyard) to LOP and GCU

Connection locations

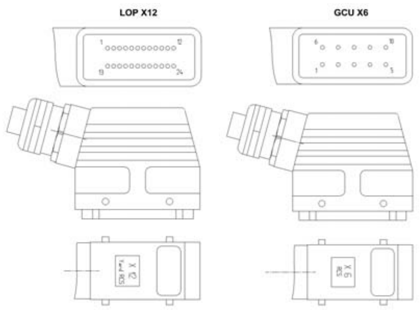
The third-party remote control is connected to connector X12 of the LOP and X6 of the GCU.

Functions

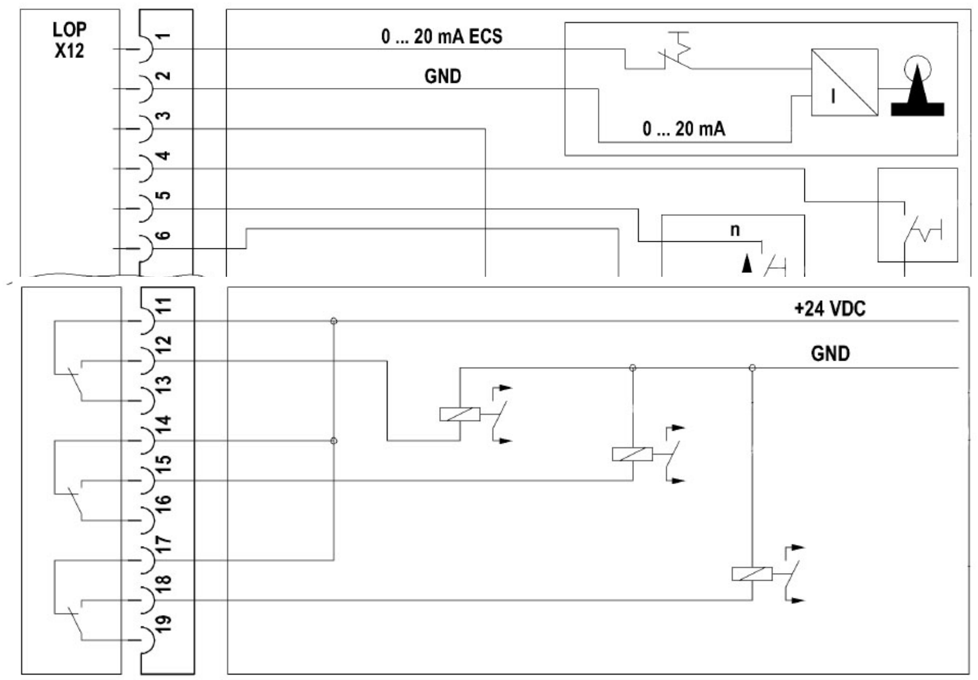
Inputs and outputs are available for the following functions:

- Engine speed input signal analog (4 mA to 20 mA), engine speed input signal – via binary signals “Increase / lower speed”, selection of speed setpoint: Binary or analog.
- Coupling specification (binary inputs).
- Disengage request from local monitoring / control to – remote control system, idling speed and status of local operation.
- Engine speed in the range in which the coupling can be actuated – (“coupling window”).
- Coupling control return signal.

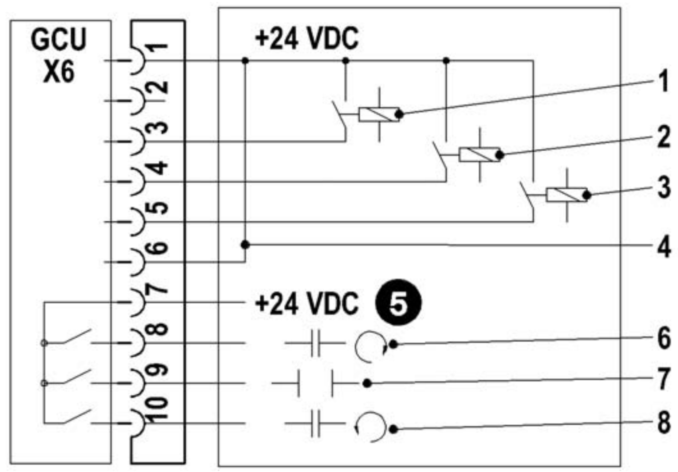
Connectors LOP X12 and GCU X6



When assembling the connector housings, ensure that the contact inserts are in the correct position.



PIN No.	Signal	Remarks
1	Input, analog speed input signal	4 to 20 mA
2	Ground	
3	Output voltage	Voltage supply +24 VDC ECS
4	Selection of speed input signal: Binary or analog.	Optocoupler input (+24VDC). Must be connected with Pin 3.
5	Binary signal, increase speed.	
6	Binary signal, lower speed.	
7	Analog voltage 0 to 10 VDC.	Proportional to engine speed.
8	GND	Ground analog
9	Analog voltage 0 to 10 VDC.	Engine load signal / injection quantity not used
10	GND	Ground analog
11	Local operation is active (potential-free relay output) max. 0.5 A.	Common
12		Make contact
13		Break contact
14	Disengagement command and / or idling speed (potential-free relay output) at Start / Stop /	Common
15		Make contact
16		Break contact
17	Emergency Stop (max. 0.5 A). Coupling window (potential-free relay output) max. 0.5 A.	Common
18		Make contact
19		Break contact



1 Input signal SYNCHRONIZATION / FLUSHING
2 Input signal NEUTRAL / DISENGAGE ENABLE / OFF
3 Input signal CONTRAROTATION / ON
4 Third-party remote control connected
5 Uext from third-party remote control
6 Return signal SYNCHRONIZATION / (ON)
7 Return signal NEUTRAL / (OFF)
8 Return signal CONTRAROTATION / FLUSHING

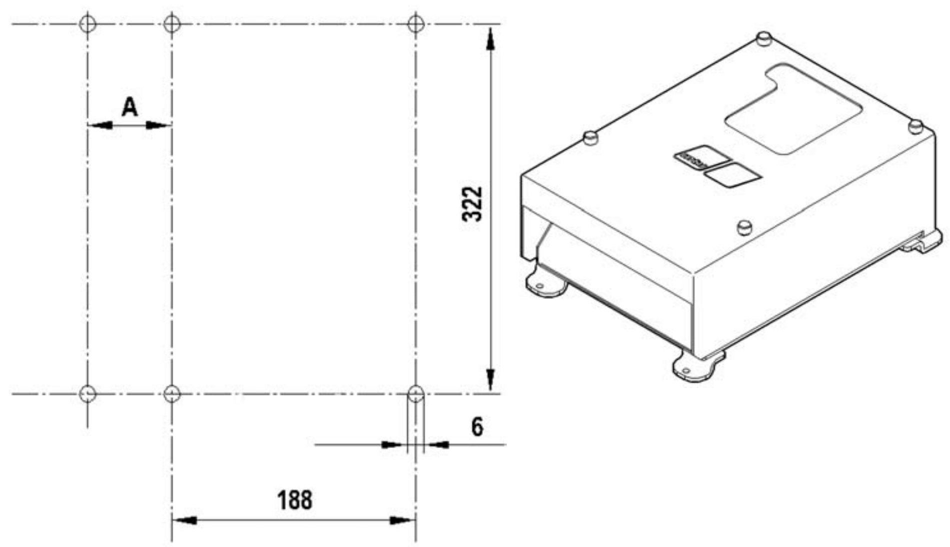
PIN No.	Signal	Remarks
1	Third-party RCS, third-party RCS is connected.	Signal +24 VDC from GCU. Must be connected to input at third-party RCS (Pin6).
2	GND	Ground
3	Input signal synchronization[1] / Flushing[2]	Must be connected with +24 VDC X6-1 (max. 2.0 A).
4	Input signal neutral[1] / disengage[2,3]	
5	Input contrarotation[1] / engage[2,3]	
6	Third-party RCS, third-party RCS is connected.	
7	Input voltage U_{ext}	U_{ext} = +24 VDC
8	Return signal synchronization[1] / engage[2,3]	+ 24 VDC, max. 0.2 A
9	Return signal neutral[1] / disengage[2,3]	+ 24 VDC, max. 0.2 A
10	Return signal contrarotation[1] / Flushing[2]	+ 24 VDC, max. 0.2 A

1 Gearbox with reversing gear FPP
2 Gearbox with reversing gear WJ (with flushing function)
3 Gearbox with reversing gear WJ / CPP

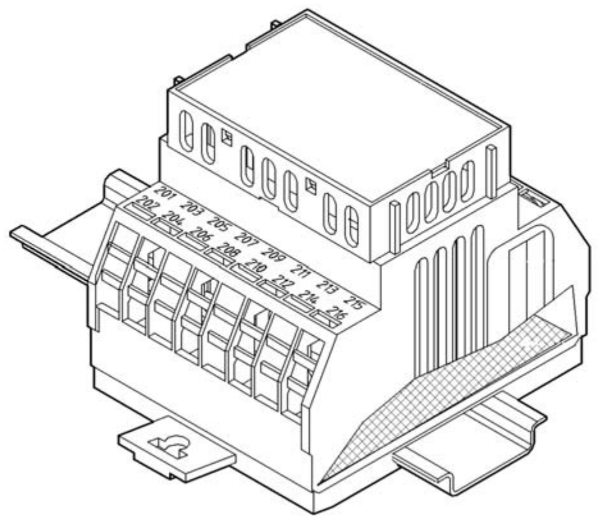
Third-party monitoring system

Mechanical installation

Installing PIM 4 module housing



Installing PIM 5 for VDR connection



The PIM 5 is installed on top-hat rails.

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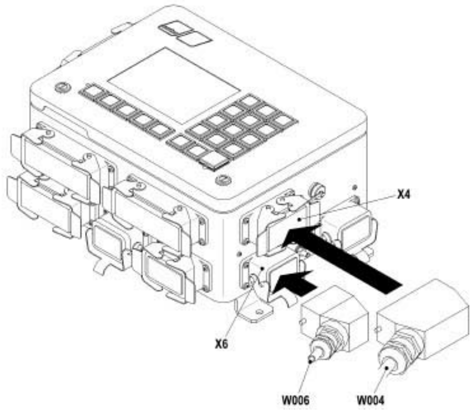
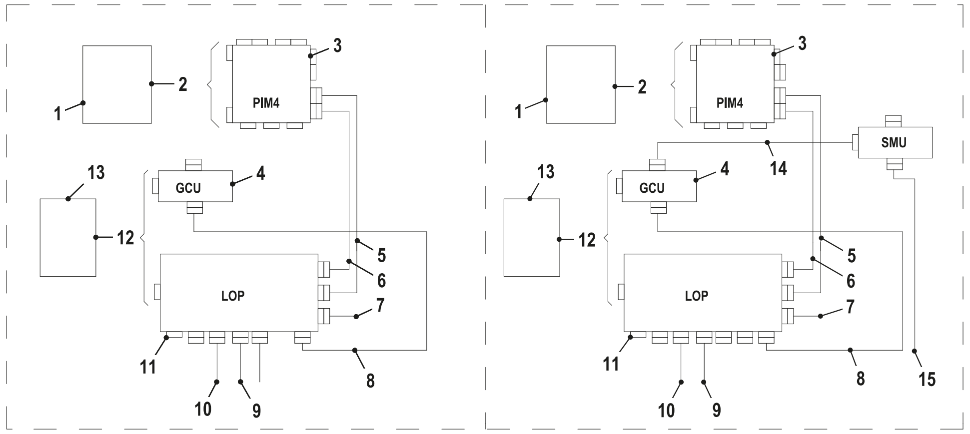
Electrical installation

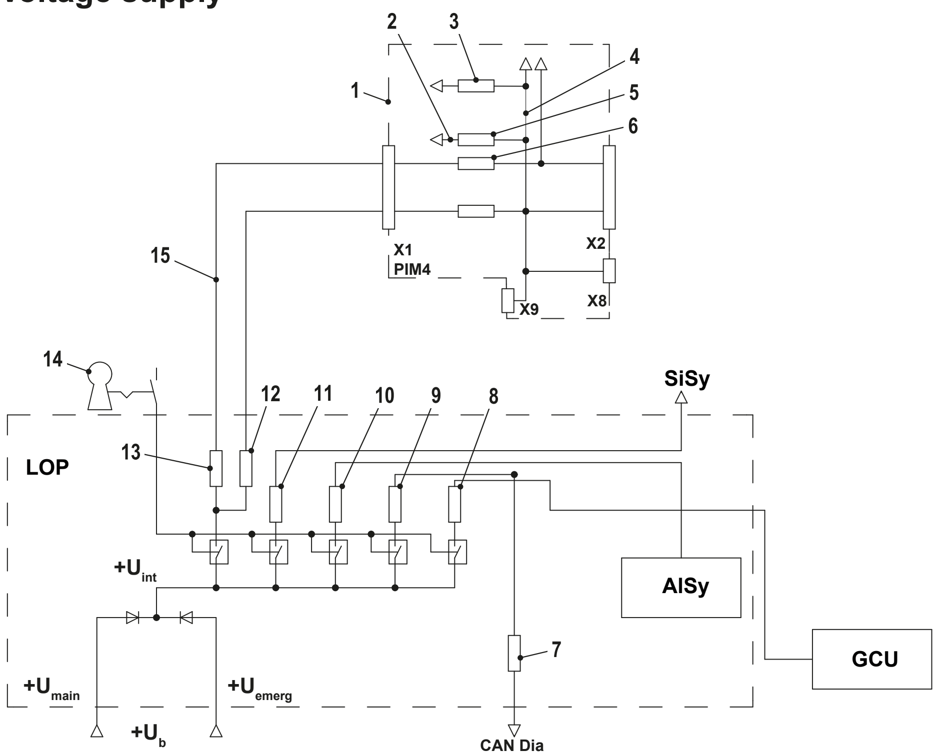
Wiring





Pos.	Designation	Meaning	Type	Supplier
1	-	Third-party monitoring system	-	Shipyard
2	Cable	Interface to third-party monitoring system	Dat, Sig	Shipyard
3	-	Control unit PIM 4	-	-
4		Connection for gearbox control GCU- gearbox	Sig	Rolls-Royce Solutions
5	Cable	Connection for control panel LOP X4 – control unit PIM 4 X3	Dat, Sig	Rolls-Royce Solutions
6	Cable	Connection for control panel LOP X6 – control unit PIM 4 X1	Volt	Rolls-Royce Solutions
7	Cable	Voltage supply, total	Volt	Rolls-Royce Solutions
8	Cable	Connection for control panel LOP – gearbox control GCU	Volt	Rolls-Royce Solutions
9	Cable	Connection for control panel LOP – engine interface EIM	Sig	Rolls-Royce Solutions
10	Cable	Control panel LOP – engine sensor "Water in fuel prefilter 1" (option: 2 cables for 2nd sensor) and option for connecting a horn and / or a flashing lamp in the engine room.	Sig	Rolls-Royce Solutions
11	Connector	For connection of additional sensors in the machine compartment (shipyard)	Sig	Rolls-Royce Solutions
12	Cable	Interface to third-party remote control system	Sig	Shipyard
13	-	Third-party remote control system	–	Shipyard
14	Cable	Connection of gearbox control GCU and shaft speed recording system SMU for plants with shaft speed recording	Volt, Sig	Rolls-Royce Solutions
15	Cable	Connection for shaft speed recording system SMU – shaft speed sensor	Sig	Rolls-Royce Solutions

Voltage supply



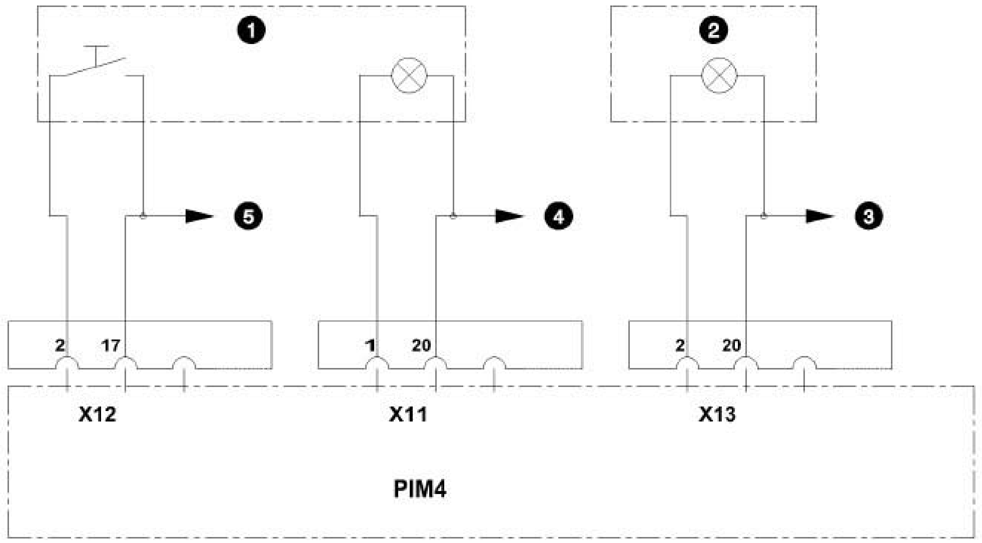
Pos.	Function name	Meaning
1		Control unit PIM 4
2	F1 on BOB 1 (1) – 3.15 A	Initialization for indicator lamps of operating buttons on control unit PIM 4.
3	F1 on BOB 1 (2) – 3.15 A	Initialization for alarm indicator lamps on control unit PIM 4.
4	$+U_{bint}$	Internal supply voltage.
5	F16 – 15 A	Input for internal supply voltage, control unit PIM 4.
6	F19 – 10 A	Input for internal supply voltage.
7	F1 – 4 A	Initialization for dialog device CAN
8	F69 – 4 A	Initialization for gearbox control GCU.
9	F67 – 20 A	Not used.
10	F70 – 15 A	Voltage supply for alarm system ALSY.
11	F71 – 3 A	Supply voltage for safety system.
12	F2 – 15 A	Output circuit for initialization of control stands (located on the pc board PSB, in LOP at the side).
13	F1 – 15 A	Output circuit for initialization of control stands (located on the pc board PSB, in LOP at the side).
14	S001	Keyswitch
15	$+U_{Out}$	Supply voltage for control unit PIM 4.

The starter is connected directly to the start battery, from where it is also initialized.
The voltage supply +Ub for all mtu devices is supplied from the LOP of the ECS.

Connection of a third-party monitoring system to the control unit PIM 4

The indicator lamps, buttons, button lamps and display instruments are connected with the connectors X10, X11, X12, X13, X14 and X18 (if available) included in the delivery to an MCS. It is advisable to divide these devices into the following groups and to connect each of these groups with a separate cable to the corresponding connector:

- Group 1: Parallel interface; Operating buttons, keyswitches, horn, button lamps, indicator lamps for prewarming (if available), engine running and ready for operation, connectors X10, X11, X12 and X18.
- Group 2: Parallel interface; all alarm indicator lamps, connector X13.
- Group 3: Parallel interface; all display instruments and their lighting, connector X14.
- Group 4: Serial interface, connector X17.



1 Start
2 OVERSPEED
3 for further alarm indicator lamps
4 for further lamps
5 for further operating buttons

Pin assignments